

Mr. ZHANG Zhenkai

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EDUCATIONAL BACKGROUND

Nanjing Tech University

09/2022-06/2026

Degree: Bachelor of Science

Major: Data Science and Big Data Technology

GPA: 88/100

The Hong Kong Polytechnic University

09/2026~

Incoming MSc Student in Data Science & Analytics

PUBLICATIONS

- ‘Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effect’, Applied Sciences, 2025. First Author. 07/2025
- ‘Behavior Prediction of Connections in Eco-Designed Thin-Walled Steel–Ply–Bamboo Structures Based on Machine Learning for Mechanical Properties’, Sustainability, 2025. Third Author. 07/2025
- ‘Exploring and Enhancing Advanced MoE Models: From Deepspeed-MoE to DeepSeek-V3’, Accepted by the IEEE 6th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT 2025), First Author 03/2025
- ‘Enhancing export potential of digital service trade in BRI countries: A stochastic frontier gravity model analysis of digital economy development and mediation pathways’, accepted by the 2025 International Conference on Digital Economy and Intelligent Computing (DEIC 2025), Third Author 03/2025
- ‘Research progress on the integration of robot vision, computer vision, and machine learning: Technological evolution, challenges, and industrial applications’, published in International Journal of Current Research in Science, Engineering & Technology (ISSN: 2581-4311), Second Author 04/2025

ACADEMIC PROJECT

Predicting Olympic medal performance for 2028: machine learning models and the impact of host and coaching effects 02/2025-07/2025

Group Project

- Classified global countries into four categories based on Olympic participation and medal outcomes, with an analysis of their 2028 medal trends
- Integrated factor analysis, time series, and four machine learning methods into the MPXG model to predict medal-winning countries medal outcomes
- Employed a FCNN to predict the probability of non-medal-winning countries winning their first medal in the next competition
- Quantified the host effect value for the past eight Olympic Games
- Developed a Bayesian linear regression-based coach effect scoring system to evaluate the extent to which coaches enhance the overall strength of athletes in a discipline

Exploring and enhancing advanced MoE models: From Deepspeed-MoE to DeepSeek-V3

01/2025-03/2025

Individual Project

- Developed a keen interest in Transformer and MoE (Mixture of Experts architecture) from *Large Language Models*
- Conducted a systematic review of MoE architecture evolution, selecting seven large language models as case studies to analyze the difference in their architectural design and optimization strategies
- Benchmarked MoE models against contemporary mainstream Transformer-based architectures, evaluating their advantage in performance and efficiency trade-offs across five key metrics

Behaviour prediction of connections in thin-walled steel-ply-bamboo structures based on machine learning for mechanical properties

07/2024-12/2025

Group Project

- Developed a thin-walled steel-ply-bamboo composite structure combining TWS and bamboo panel advantages
- Conducted factor analysis and data processing to evaluate optimal mechanical properties of TWS structure connections
- Implemented eight machine learning models for connection classification, identifying the top-performing model
- Applied five hyperparameter optimization algorithms to enhance model performance/efficiency and mitigate overfitting

HONORS

Meritorious Winner (Problem F), Mathematical Contest in Modeling (MCM)

2/2026